Midterm — Final Project Idea Presentation

61FIT4MLA Machine Learning Applications | Week 6 | 30% of the course grade

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What this is

Your midterm is a **group presentation of your final-project idea**. There is no written midterm examination this semester.

You present the problem you intend to solve, the data you intend to use, the machine-learning task you have formulated, and how you will evaluate it. You are graded on the *quality of the plan and your reasoning*, not on results — you are not expected to have trained anything yet.

The theme of the course is **educational support**: machine learning that helps a teacher or a learner. Your project sits inside that theme. How you enter it is up to your group:

- **Learner-response data** — predicting whether a student answers a question correctly, which topics a learner is weak in, which questions discriminate well. This is the closest match to the labs you have been doing all semester, and the easiest route to a solid project.
- **Translation and language-learning support** — error typing, feedback prioritisation, difficulty estimation on language data. A natural fit at HANU, and the instructor's own research direction.
- **Another educational-support problem** you can find real data for. Bring it to the instructor before Week 4.

None of these is worth more marks than another. Choose the one whose data you can actually obtain.

The framing rule that applies to every project

Whatever you build is a **support tool for a human teacher or learner**. It is not an automatic grader, not an AI teacher, and not a replacement for human judgement. The human remains the final decision-maker, and every model output is a suggestion to be checked.

A presentation that proposes to replace the teacher will lose marks under criterion 5, however good the machine learning is.

Format

Group size	as registered for the final project
Presentation	**7 minutes**, strictly timed
Questions	**3 minutes** from the panel and the class
Slides	**6–8 slides**, PDF only
Speaker	**one** member presents for the group
Submission	**one representative** uploads the PDF, **before 20:00 the day before**

Slides are collected the night before and projected from one machine. This keeps the session to time and removes laptop-swapping. A group that has not uploaded by the deadline presents last and loses the delivery marks.

What your slides must cover

Six things. One slide each is a reasonable plan.

1. **Problem and user.** What educational-support problem are you addressing? *Who* is the human being supported — a teacher deciding who needs help, a learner practising, a curriculum designer? What decision does your tool inform?

2. **Data.** Which data route (public benchmark, student-curated mini-dataset, instructor-provided, or faculty collaboration)? Name the specific dataset. State its size, its fields, and its **licence**. If you do not yet have the data in hand, say exactly what you will do to get it and by when. Learner-response tables of the kind used in the weekly labs are a legitimate and well-supported choice.
3. **ML task alignment.** Turn the problem into a machine-learning task. State the input, the output, and what one training example looks like. If your labels are subjective or imbalanced, say so now.
4. **Model plan.** Name at least **two model families from this course** you intend to apply, and say *why those two* for this task. "We will try everything" is not a plan.
5. **Evaluation plan.** How will you split the data? Which metrics, and why is accuracy alone insufficient here? What is your **baseline** — the simple thing your model must beat to be worth anything?
6. **Risks and responsible use.** What is most likely to go wrong? Where could the tool mislead a teacher or a learner? What stays under human control?

How you are graded

The midterm mark is out of 100 and is split evenly:

Component	Share of the midterm mark
Presentation of your final-project idea (group mark)	50%
Completion of Lab 1 to Lab 5 (individual mark)	50%

Presentation mark. Graded out of 100 against the criteria below.

#	Criterion	Marks
1	Problem formulation, and a clearly identified human user	20
2	Data: route, specific dataset, licence, feasibility	20
3	ML task alignment and justified choice of two model families	25
4	Evaluation plan, split discipline, metrics, and a named baseline	15
5	Risks, limitations, and responsible human-in-the-loop framing	10
6	Delivery: timing, clarity, and answers to questions	10
	Total	**100**

Only one member speaks, and the whole group receives the same presentation mark. Every member should still contribute to the plan; the panel may ask any question about the project.

Lab completion: half of the midterm mark

Lab mark. Each of Lab 1 to Lab 5 is worth 20 out of 100. You earn the 20 marks for a lab when you submit it by its posted deadline. Completing all five labs gives the full 100 for this component.

This is a completion check, not a quality check: a submitted, genuine attempt counts. The lab mark is **individual**. It depends only on your own submissions, not on those of your teammates.

Your midterm mark is

$$\text{midterm} = 0.5 \times \text{presentation mark} + 0.5 \times \text{lab mark}.$$

Example. Your group scores 80 on the presentation. If you completed all five labs, your midterm mark is $0.5 \times 80 + 0.5 \times 100 = 90$. A teammate who completed three labs (lab mark 60) receives $0.5 \times 80 + 0.5 \times 60 = 70$.

The midterm mark out of 100 is scaled to the 30% midterm component of the course grade. There is no bonus and no cap: the two halves add up to the 100 marks.

Deadlines

Group registration and provisional title	**end of Week 4**
Lab 1–5 submissions (half of the midterm mark)	**their own posted deadlines**
Slides uploaded as PDF	**20:00, the day before Week 6**
Presentations	**Week 6, in class**

What you do not need yet

No trained model. No results table. No final dataset if you have documented how you will obtain it. No language-model component — that belongs to the optional research track and is not required.

What you *do* need is a problem a human actually has, data you can really get, a task you can really formulate, and an honest account of what might fail.

Advice from previous cohorts

The three ways groups lose most marks:

- **A dataset with no task.** "We found an interesting corpus" is not a machine-learning project. Criterion 3 is the largest single block of marks.
- **No baseline.** If you cannot say what simple method your model must beat, you cannot later claim it worked.
- **Running over time.** Seven minutes is short. Rehearse it. The panel stops you at seven.